Redefining High-Performance Computing in Compact Spaces

The COM-HPC Mini packs powerful computing into compact devices like robots, drones, and industrial automation systems, ensuring top-tier performance in tight spaces.

The COM-HPC Mini introduces a new specification within the COM-HPC ecosystem, aimed at enhancing high-performance computing in a much smaller footprint. As part of the broader COM-HPC family, which includes the high-performance standards for client and server modules, COM-HPC Mini focuses on delivering top-tier performance in a compact size of just 95mm×70mm. This smaller form factor makes it ideal for applications where space is at a premium but high performance is non-negotiable, such as DIN-rail PCs in industrial automation or portable test and measurement devices.

Despite its diminutive size—nearly half that of its closest COM-HPC relative—the Mini variant does not compromise on capabilities. It offers a robust solution for autonomous mobile robots, drones, and mobile 5G devices, which require efficient, high-performance modules within a constrained space. The design also features a reduced stack height of 15mm from the carrier board to the top of the module, a decrease from other variants by 5mm. This necessitates the use of soldered memory to boost durability and enhance thermal coupling to heat spreaders, ensuring performance amidst physical and thermal challenges.

As the technology landscape evolves, the demand for smaller yet powerful computing solutions is becoming increasingly crucial. The PICMG (PCI Industrial Computer Manufacturers Group) has responded by developing COM-HPC Mini. This new standard not only continues the legacy of the original COM-HPC modules, but also complements smaller form factors like COM Express Mini, providing a significant upgrade in performance and interface capabilities for space- and power-limited applications. As we move forward, COM-HPC Mini is set to redefine the potential of small form factor computing modules, pushing the

COM-HPC module classes

Feature	COM-HPC Server Module	COM-HPC Client Module	COM-HPC Mini Module
Application	Edge computing in harsh environments such as autonomous vehicles, cell tower base stations	High-end embedded applications like medical equipment, industrial systems, casino gaming, rugged PCs	Small form factor applications needing high-end computing power
Display requirements	Headless (no display required)	Requires one or more displays	N/A (not specified)
Memory capacity	Up to eight full-size DIMM slots	Up to four SO-DIMM slots	Not specified but uses a single 400-pin connector
I/O capabilities	Multiple high-speed Ethernet ports (10Gbps or 25Gbps), up to 65 PCIe lanes	49 PCIe lanes	USB 4.0, Thunderbolt, PCIe Gen 5, 10Gbps Ethernet
Input power	Minimum 358W; compatible with CPUs dissipating up to 150W	Up to 251W at the lower end of an 8V power supply, more with stable 12V supply	Up to 107W at the lower end of an 8V spectrum
Design considerations	Robust, durable, and upgradable for tough conditions	Geared towards applications demanding high graphical performance and advanced I/O bandwidths	Compact, power-efficient design for high performance in credit-card-sized package
Unique attributes	Excels in CPU performance and extensive I/O bandwidth, perfect for harsh environments	Ideal for applications needing displays and varied I/O options	Performance-oriented despite space, power, or cost constraints